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The Deployment Problem: Substrate Lifecycle as Distributed State at the C5ISR Edge (Series Capstone)

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Abstract

The preceding ten papers in this series have specified a tactical substrate of considerable architectural sophistication: a DTN transport layer that survives link degradation, a write-ahead log that maintains append-only state integrity across disconnected nodes, a CRDT coherence layer that converges divergent state across connectivity windows, an AI Supervisor that operates within a human-defined governance envelope, and three governance-layer architectures — classification integrity, accreditation integrity, and coalition interoperability — that maintain the security and governance properties the substrate must exhibit in operational use. What none of those papers has addressed is how this substrate arrives at an operational node. How it is verified to match the authorized baseline. How it is updated when update is necessary. How it survives an update that partially fails in a DDIL window. How the AI Supervisor model is maintained as the operational envelope evolves. And how the operational team knows, with the same cryptographic confidence that the WAL provides for data integrity, that the substrate they are relying on is the substrate that was authorized, tested, and built to operate at the tactical edge.

This paper argues that the deployment problem in DDIL is not an engineering problem — it is a governance problem about what "deployed" means when the deployment environment is the same degraded edge the substrate was designed to survive. Enterprise deployment models assume connectivity, sequenced rollout, rollback capability, and stable targets. The tactical edge provides none of these reliably. A deployment architecture designed for enterprise assumptions will fail at the tactical edge not by deploying incorrectly — it may deploy correctly — but by deploying to an environment where its operational assumptions (network-accessible package repositories, immediate health-check feedback, reliable rollback channels) are violated at exactly the moment when deployment events are most likely to occur: during port calls, during connectivity windows, at operational tempo.

The paper's central architectural claim is that fielding — the tactical edge's version of deployment — must treat the substrate itself as a distributed state management problem. Updates are bundles, not package installations. Rollback is a CRDT merge operation, not a filesystem restoration. AI Supervisor model updates are WAL events, not configuration file swaps. The deployment state of every node is a first-class WAL artifact, visible to the governance infrastructure with the same precision as the classification state and accreditation state the preceding papers specified. And the deployment lifecycle — from authorized build through fielding through in-service update through decommission — is governed by the H half before the AE^2 half can enforce it.

How to cite

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ABSTRACT

The Edge Degraded Operations Undecalogy has, across its first ten papers, specified the substrate over which agentic operation must function under DDIL conditions: the tactical substrate's four properties (P1), HGC³AE² translated to the degraded edge (P2), variable-link communication via DTN bundle custody (P3), state coherence under partition via WAL + CRDT (P4), out-of-path AI supervision (P5), PACE-plan protocol selection (P6), Coherence-SLA workload classification (P7), distributed-state classification integrity (P8), continuous-validity accreditation (P9), and coalition governance translation (P10). This eleventh paper is the **capstone**: a synthesis that treats the **substrate lifecycle itself** — provisioning, operation, refresh, retirement — as distributed state, and specifies how the substrate's lifecycle integrates the ten preceding papers into a coherent architecture for governed agentic operation on the tactical edge.

Closure of this capstone unblocks downstream inheritance at G3 citation-fixture readiness for the Edge Degraded Operations series.

HOW TO CITE

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1. INTRODUCTION

The Edge Degraded Operations Undecalogy's first ten papers each specify an architectural commitment that the tactical substrate must satisfy. What that series has not yet done is integrate the ten commitments into a coherent specification of the substrate's lifecycle — provisioning, operation, refresh, retirement — as a single distributed-state problem. The present paper does that integration.

The substrate's lifecycle is itself distributed state. A forward-deployed cognitive node is provisioned with a baseline configuration, operates for months or years under DDIL conditions, accumulates persistent memory and operational history, may be refreshed in-place or replaced by a newer-generation node, and is eventually retired with its accumulated state either preserved or sanitized. Each stage is a governance-significant event; the lifecycle as a whole is the architecture's continuity property writ large.[1,2]

The paper proceeds as a specification-shaped capstone. §2 walks the substrate's lifecycle stages against the ten preceding EDO papers. §3 presents the cross-reference matrix. §4 specifies conformance criteria. §5 surfaces operational findings. §6 returns to HGC³AE² and closes the Undecalogy.

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1 Justin H. Kuiper, CISSP, *HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

2 Justin H. Kuiper, CISSP, *Alistair Prime in a Box* (v1.0-preprint, 2026), zenodo.org/records/19869307.

2. SUBSTRATE LIFECYCLE STAGES

2.1 PROVISIONING

The substrate is provisioned with: baseline configuration (EDO-P9 accreditation evidence), classification metadata (EDO-P8), pre-authorized governance envelope (EDO-P2 H translation), initial persona registry (MAO-P7 in cloud-native form, adapted here for tactical substrate), substrate-class declaration (EDO-P7), and PACE-plan tier specifications (EDO-P6).[3,4,5,6,7]

2.2 OPERATION

The substrate operates: DTN custody for variable-link cargo (EDO-P3); WAL + CRDT for state coherence under partition (EDO-P4); out-of-path supervision for orchestration assurance (EDO-P5); continuous-validity accreditation as runtime property (EDO-P9); coalition translation at coalition boundaries (EDO-P10); classification integrity at substrate boundaries (EDO-P8); Coherence-SLA enforcement per workload class (EDO-P7); PACE-plan tier transitions under degradation (EDO-P6).[8,9,10,11]

2.3 REFRESH

The substrate is refreshed: hardware refresh, accelerator generation rollover, software updates, fine-tune deployments, persona-harness updates. Each refresh is a re-accreditation trigger (EDO-P9), may alter classification metadata (EDO-P8), and propagates through the governance envelope (EDO-P2).

2.4 RETIREMENT

The substrate is retired: persistent state is either preserved (transferred to a successor substrate with classification preserved) or sanitized (cleared with audit trail). Retirement is governance-authorized; the authority chain is explicit (EDO-P9).

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3 EDO-P2 promotion (in concurrent drafting; yks-build#104).

4 EDO-P6 promotion (in concurrent drafting; yks-build#105).

5 EDO-P7 promotion (in concurrent drafting; yks-build#106).

6 EDO-P8 promotion (in concurrent drafting; yks-build#107).

7 EDO-P9 promotion (in concurrent drafting; yks-build#108).

8 EDO-P3 promotion (in concurrent drafting; yks-build#95).

9 EDO-P4 promotion (in concurrent drafting; yks-build#96).

10 EDO-P5 promotion (in concurrent drafting; yks-build#97).

11 EDO-P10 promotion (in concurrent drafting; yks-build#109).

3. CROSS-REFERENCE MATRIX

Lifecycle Stage Element

EDO Paper

Tactical substrate four properties (DDIL, degraded compute, partial governance, adversarial pressure)

EDO-P1

HGC^3AE^2 letter-by-letter translation to degraded edge

EDO-P2

DTN bundle custody (BP v7 + BPSec)

EDO-P3

WAL + CRDT for state coherence

EDO-P4

Out-of-path AI supervisor (monitor != monitoree)

EDO-P5

PACE-plan protocol selection (Primary/Alternate/Contingency/Emergency)

EDO-P6

Coherence-SLA workload classification

EDO-P7

Distributed-state classification integrity

EDO-P8

Continuous-validity accreditation

EDO-P9

Coalition governance translation

EDO-P10

Substrate lifecycle synthesis (this paper)

EDO-P11

Mission-Ready Decalogy inheritance: P1 HGC³AE² Framework · P3 Edge AI Doctrine · P4 Alistair Prime in a Box · P5 Skipjack Protocol. Managing Agentics Ops Decalogy cross-link: MAO-P9 Security & Provenance.[12][13]

4. CONFORMANCE SPECIFICATION

An organization may claim conformance to the EDO Undecalogy by demonstrating: tactical-substrate four-property awareness (P1); HGC³AE² translated implementation (P2); DTN bundle custody with BP v7 + BPSec (P3); WAL + CRDT persistence (P4); out-of-path supervision (P5); PACE-plan tiered protocols (P6); Coherence-SLA classification (P7); distributed-state classification integrity (P8); continuous-validity accreditation (P9); coalition translation discipline (P10); and substrate lifecycle management with explicit stage handling (P11).

Each commitment is auditable through the cross-reference matrix; the architecture is decidable.

5. OPERATIONAL FINDINGS

Three findings load-bearing for adopters.

Lifecycle stages are sources of governance events. Every transition between provisioning, operation, refresh, and retirement is a governance event with implications for accreditation, classification, and envelope authority. Treating lifecycle as routine operations loses governance significance.

Refresh is the under-engineered stage. Most operators have invested in provisioning and operation; refresh is the stage where most failure modes recur (incomplete re-accreditation, classification drift, envelope ambiguity, harness-version drift).

Retirement is the silent stage. Sanitization discipline, state transfer to successor substrates, and audit-trail preservation at retirement are routinely under-engineered. Adversaries take advantage; spillage at retirement is a recurring incident pattern.

12 Mission-Ready AI Decalogy (MR-P1 through MR-P10).

13 Managing Agentics Ops Decalogy (MAO-P1 through MAO-P10).

6. IMPLICATIONS AND CONCLUSION

The implications of the substrate-lifecycle-as-distributed-state synthesis extend to organizational practice (lifecycle stages are governance ceremonies, not infrastructure-team events), architecture (each lifecycle transition requires explicit architectural support), and adoption sequencing (organizations adopt the EDO Undecalogy in lifecycle-aligned phases).

The core claim, closing the Undecalogy: **the tactical substrate's lifecycle is itself distributed state requiring the same architectural commitments as the operational state running on it; the ten preceding EDO papers each specify a commitment that the lifecycle must satisfy, and the capstone integrates them**.

Returned to HGC^3AE^2 letter by letter, integrating across the Undecalogy:

H — Human-governed. Governance authority is preserved through pre-authorized envelopes that survive disconnection (EDO-P2); accreditation authority is engaged at lifecycle transitions (EDO-P9); coalition translation rules are governance-authored (EDO-P10); substrate retirement is governance-authorized (this paper).

G — Curated Context. Persistence anchored locally with CRDT merge discipline (EDO-P4); classification metadata as state property (EDO-P8); coalition regime registry (EDO-P10); substrate baseline-as-runtime-property (EDO-P9).

C^1 — Cybersecurity. Adversarial pressure assumed on every layer (EDO-P1); BPSec at every custody point (EDO-P3); classification integrity (EDO-P8); attestation for security (cross-link MAO-P9).

C^2 — Continuity. The Undecalogy's primary commitment; preserved through pre-authorized envelopes, WAL durability, CRDT merge, PACE-plan tiered protocols, out-of-path supervision, continuous-ATO, and substrate lifecycle management.

C^3 — Coherence. Maintained through CRDT merge discipline (EDO-P4), Coherence-SLA classification (EDO-P7), and coalition translation reversal discipline (EDO-P10).

A — Agentically Engineered. Within pre-authorized envelopes under disconnection (EDO-P2); within Coherence-SLA per class (EDO-P7); within accreditation baseline (EDO-P9); within classification authority (EDO-P8).

E — Executed. Continuous execution under DDIL conditions across the lifecycle; out-of-path supervised; PACE-plan tier-appropriate.

The exponent — the closure — is the lifecycle-event-to-governance-update loop. Every lifecycle transition feeds governance ceremony; governance updates propagate to substrate; the substrate executes the updated envelope.

This paper closes the Edge Degraded Operations Undecalogy and unblocks downstream inheritance at G3 citation-fixture readiness. The Undecalogy is parallel to the Mission-Ready Decalogy and complementary to the Managing Agentics Ops Decalogy; together the three series specify the architecture, the operational layer, and the degraded-conditions adaptation of governed agentic operation.[14][15][16]

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14 Forthcoming MR-P10 *Reference Architecture* CAPSTONE (in concurrent drafting; yks-build#90).

15 Forthcoming MAO-P10 *Robo Stack in Production* CAPSTONE (in concurrent drafting; yks-build#91).

REFERENCES

— end v1.0-preprint-draft —

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16 Robert K. Yin, *Case Study Research and Applications: Design and Methods*, 6th ed. (Thousand Oaks, CA: Sage, 2018).